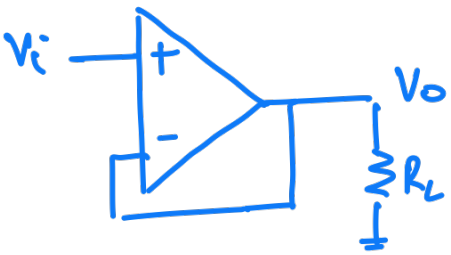
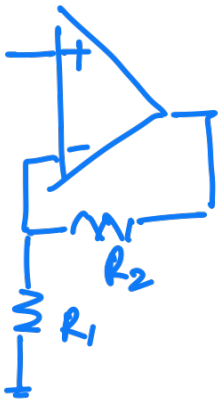
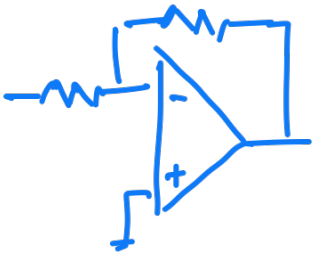
Lecture #08

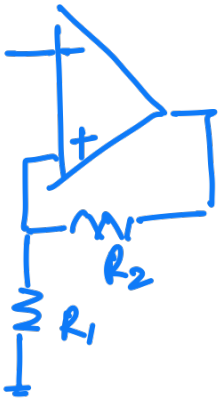


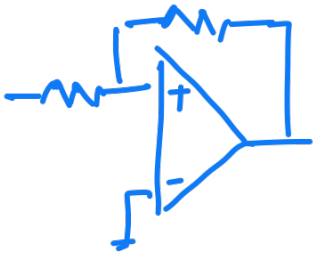
Voltag-buffer

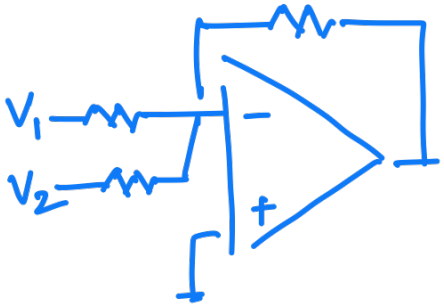


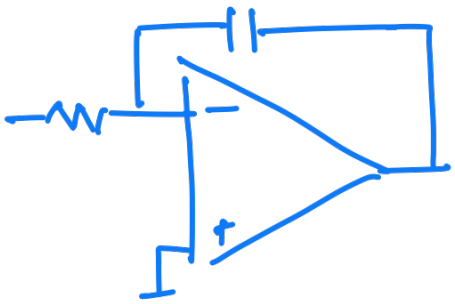


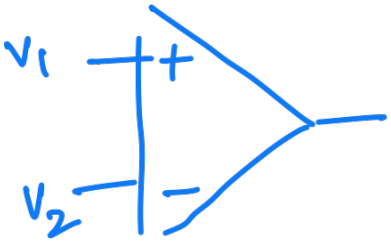
"Schmitt Trigger"

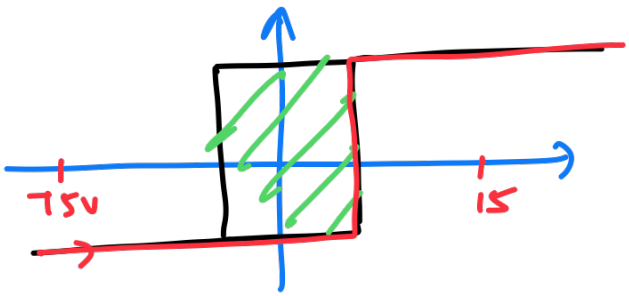


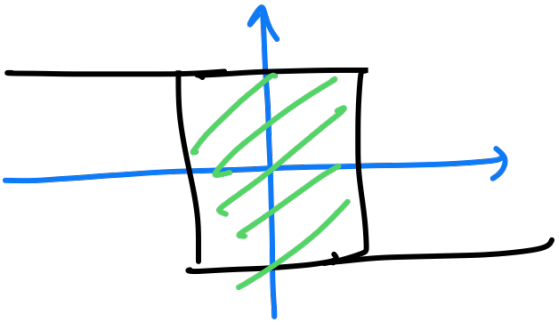


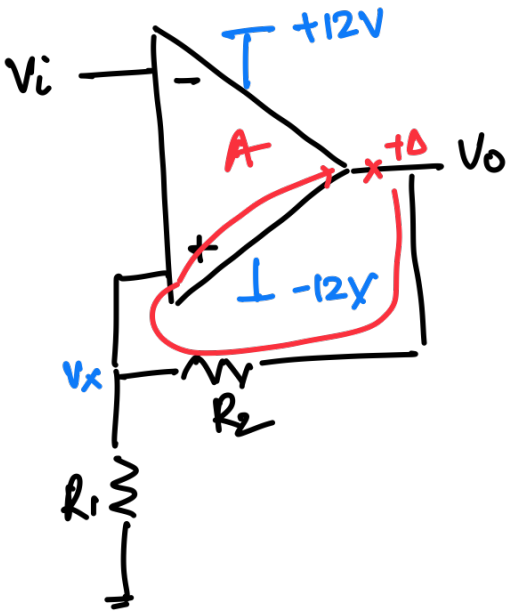












$V_i = +100V$

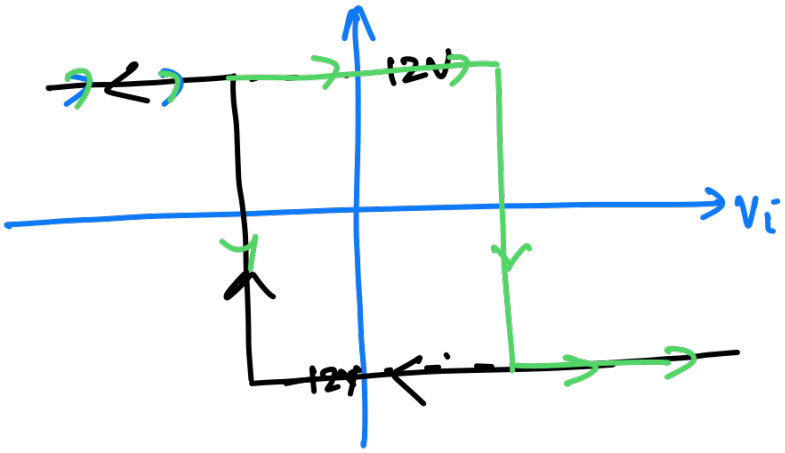
#1 $V_o = +12V \Rightarrow V_x = \frac{R_1}{R_1+R_2} * 12V$

$V_e = \frac{R_1}{R_1+R_2} * 12 - 100V < 0$

$V_o = -12V$,

$V_e = \frac{-R_1}{R_1+R_2} \times 12 - V_i > 0$

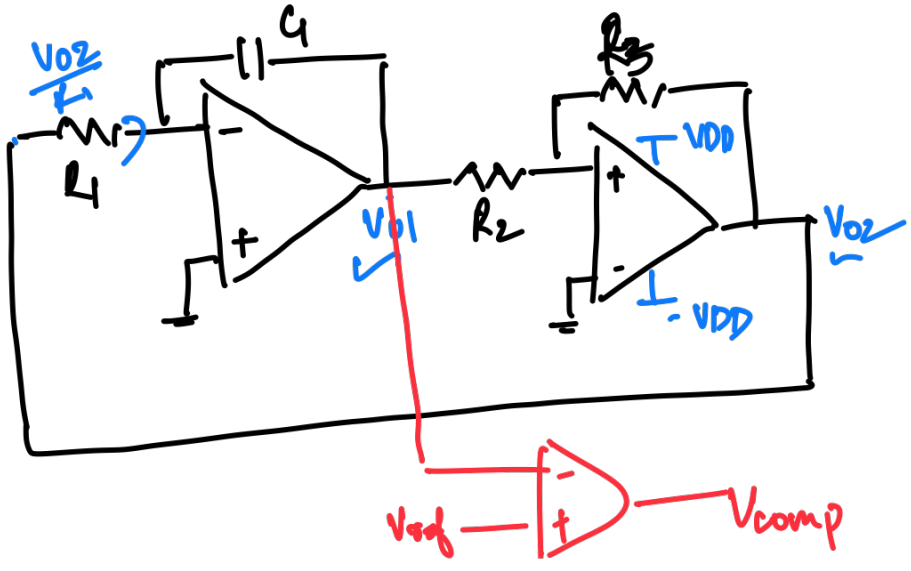
$V_i < \frac{-R_1}{R_1+R_2} \times 12V$

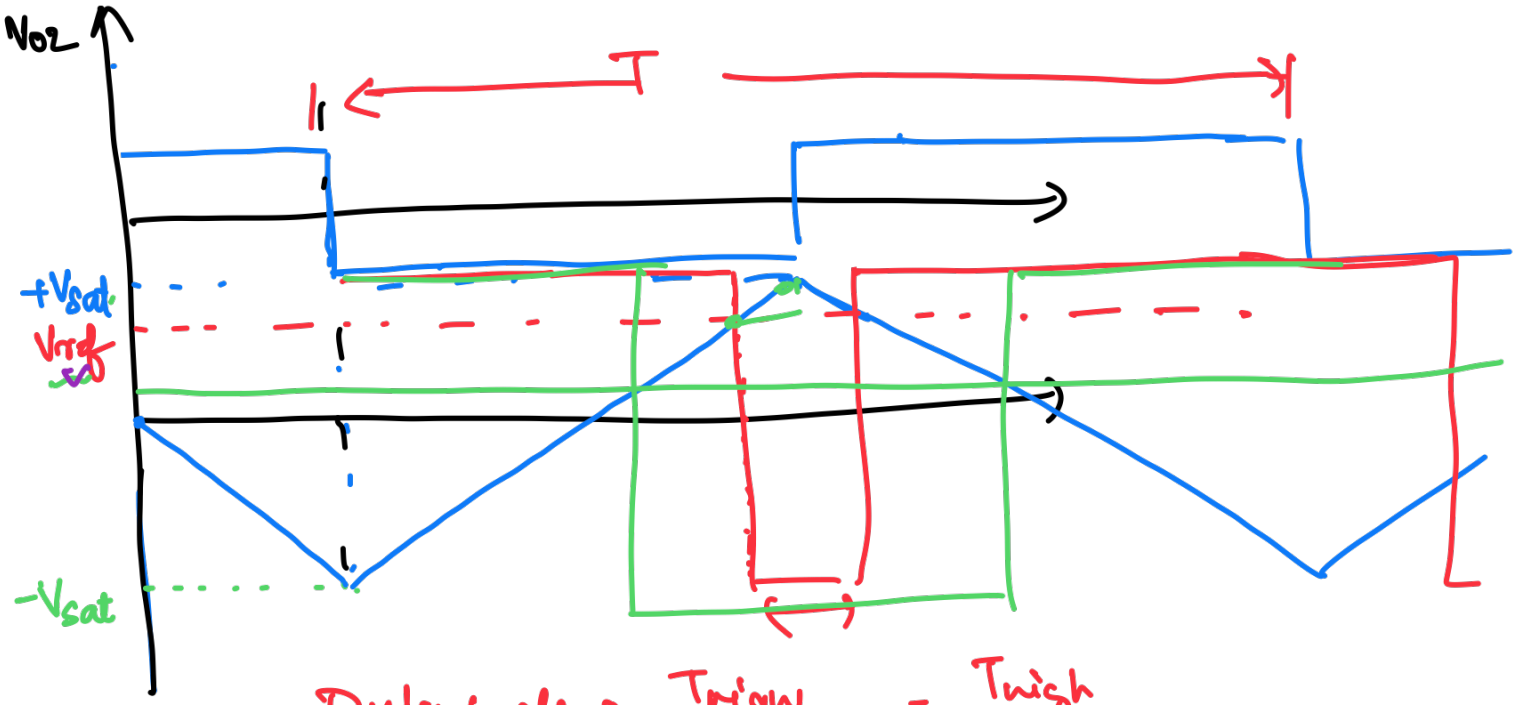


$V_e = +\frac{R_1}{R_1+R_2} \times 12V - V_i < 0$

$+|V_i|$

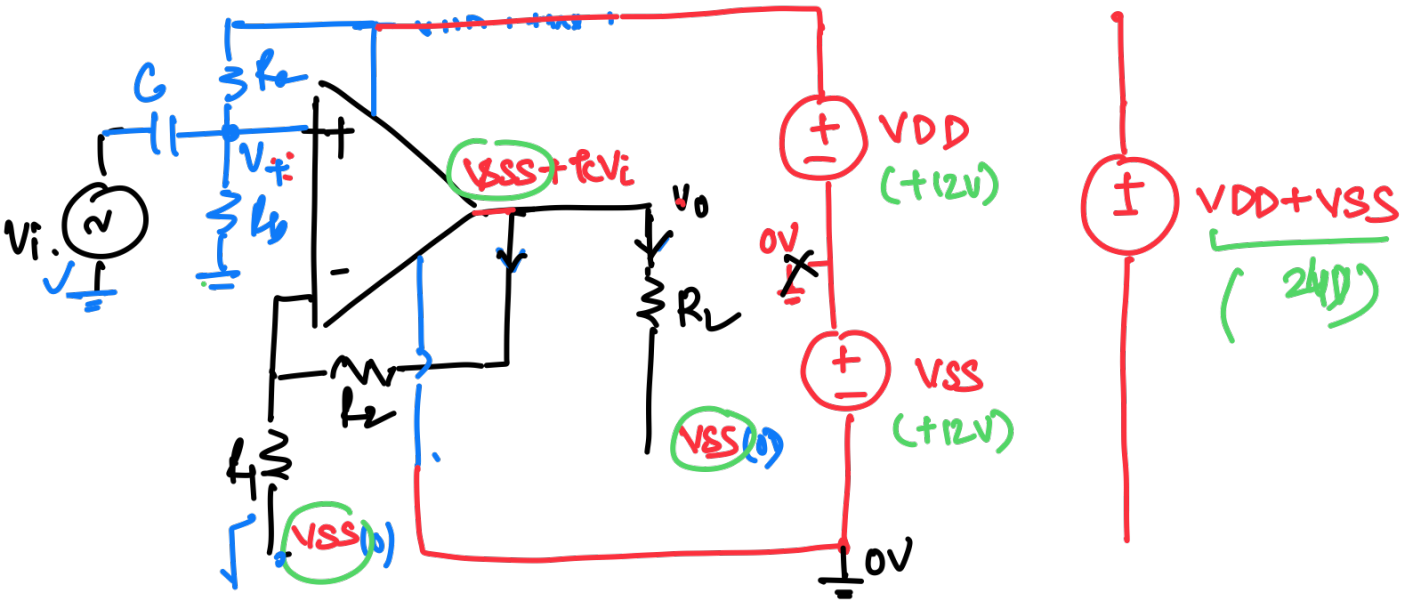
$V_i > \frac{R_1}{R_1+R_2} \times 12V$



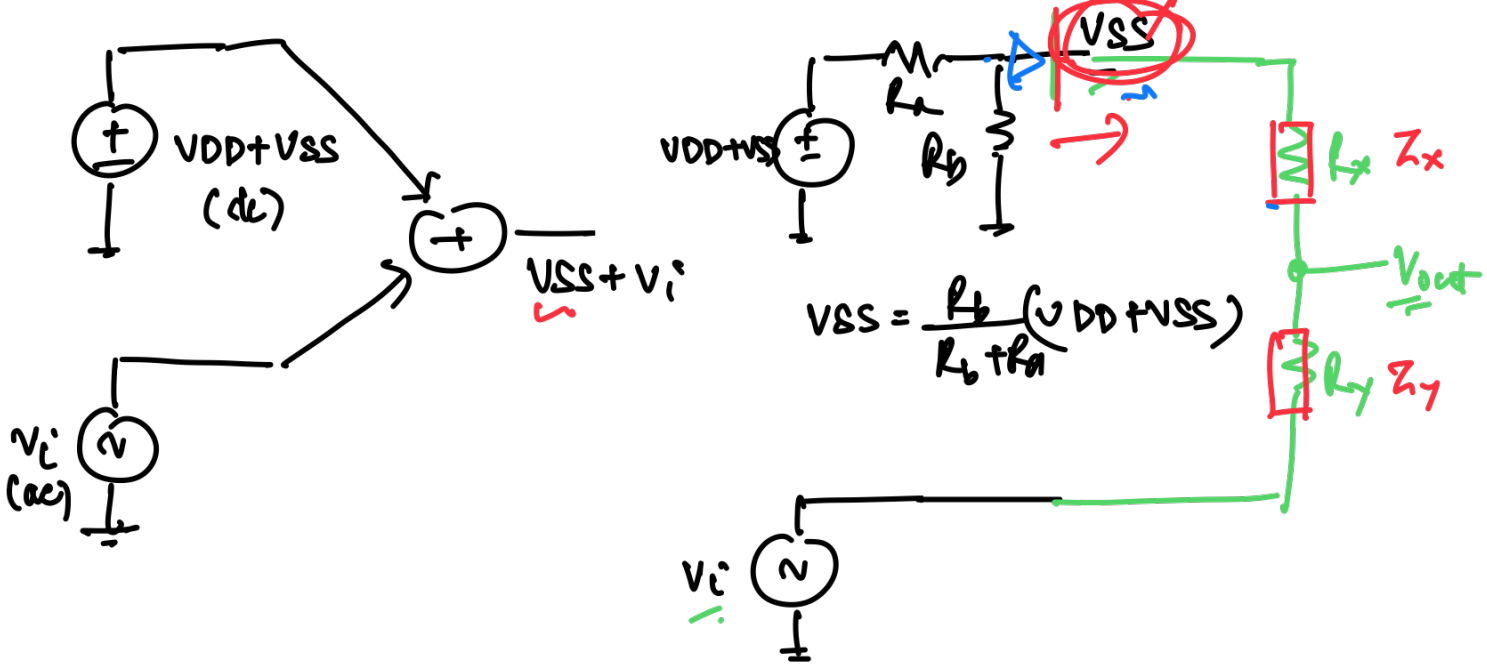


$$\text{Duty cycle } \eta = \frac{T_{high}}{T_{high} + T_{low}} = \frac{T_{high}}{T}$$

Pulse-width Modulated Waveform (PWM)



$$\frac{V_o}{V_i} = \left(1 + R_2/R_1\right) = k$$



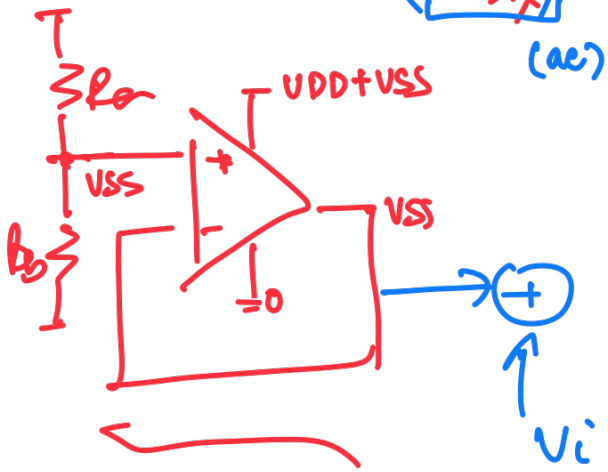
$$V_{out} = \frac{R_x}{R_x+R_y} V_i + \frac{R_y}{R_y+R_x} V_{SS}$$

$$= \frac{Z_x}{Z_x+Z_y} V_i + \frac{Z_y}{Z_y+Z_x} V_{SS}$$

1 → ac, 0 → dc (1 → dc, 0 → ac.)

$$= \left(\frac{1}{1+\frac{Z_y}{Z_x}}\right) V_i + \left(\frac{Z_y \| Z_x}{\frac{Z_y}{Z_x}+1}\right) V_{SS}$$

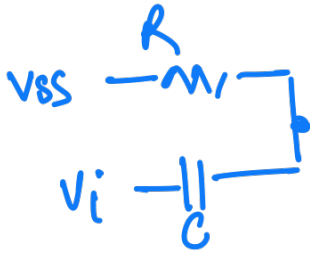
(ac) (dc)



$$\frac{Z_y}{Z_x}\Big|_{ac,} \to 0$$

$$\frac{Z_y}{Z_x}\Big|_{dc} \to \infty$$

$$\frac{1}{sC.R}\Big|_{s\to 0} \to \infty$$



$$V_{out} = \frac{R}{R+\frac{1}{sC}} V_i + \frac{\frac{1}{sC}}{\frac{1}{sC}+R} V_{SS}$$

$$= \frac{sCR}{1+sRC} V_i$$

$|sRC| > 10$

$|w_{in}RC| > 10$

$$c > \frac{10}{w_{in} \cdot L}$$